



**DEEPPFAKE IMAGE DETECTION USING
COLOR CHANNEL CO-OCCURRENCE
MATRICES AND CONVOLUTIONAL
NEURAL NETWORKS**



**MINI PROJECT REPORT
(23EXC302)**

Submitted by

VELMURUGAN U

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MASTER OF COMPUTER APPLICATIONS

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KARPAGAM COLLEGE OF ENGINEERING
COIMBATORE – 641 032
ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled “**DEEPFAKE IMAGE DETECTION USING COLOR CHANNEL CO-OCCURRENCE MATRICES AND CONVOLUTIONAL NEURAL NETWORKS**” is the Bonafide work of **VELMURUGAN U (717824Z154)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

Signature of the HoD with date

Dr. P. SHANTHI

Professor & HoD

School of Computer Applications

Karpagam College of Engineering

Coimbatore – 641 032

Signature of the Supervisor with date

Dr. P. SHANTHI

Professor & HoD

School of Computer Applications

Karpagam College of Engineering

Coimbatore – 641 032

Certified that the candidate was examined during the viva voce examination held on _____

Signature of the Internal Examiner

with date

Signature of the External Examiner

with date

SCHOOL OF COMPUTER APPLICATIONS

VISION

To be a nationally prominent and recognized department in academics and research activities with the aim of developing competent professionals to serve the society in the ever-changing industry.

MISSION

- Enabling the students to be knowledgeable and creative by developing state of the art curriculum and innovative teaching methods.
- Producing highly competent professionals by providing training programs to bridge the gap between the institution and industry.
- Enhancing the quality of research and productivity by providing the required facilities.

PROGRAMME EDUCATIONAL OBJECTIVES



- **PEO 1:** Graduates will be competent enough to identify, analyse, design and develop software solutions to the problems in the field.
- **PEO 2:** Graduates, equipped with good communication skills and technical expertise will be able to conceptualize and lead projects with an aim to become effective entrepreneurs or employees.
- **PEO 3:** Graduates, with exposure to highly active research environment, will have the motivation to participate and contribute in research and development.

PROGRAMME OUTCOMES (POS)

STUDENTS IN THE MCA PROGRAMME WILL HAVE THE ABILITY TO

1. Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
2. Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
3. Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
4. Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
6. Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.
7. Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

8. Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
9. Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
10. Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practices.
11. Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments.
12. Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

DECLARATION



I hereby declare that this Project report entitled “**DEEPFAKE IMAGE DETECTION USING COLOR CHANNEL CO-OCCURENCE MATRICES AND CONVOLUTIONAL NEURAL NETWORKS**” submitted by me for the degree of **Master Of Computer Applications at Karpagam College of Engineering, Coimbatore** is the record of original work done by me under the guidance and supervision of **Dr. P. SHANTHI, Professor** at the School of Computer Applications, Karpagam College of Engineering, Coimbatore –641032 and has not formed the basis for the award of any degree, or diploma or titles in this Institution or any other Institution of higher learning.

(Signature of the Candidate)

VELMURUGAN U

717824Z154

Date:

Place: Coimbatore

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VELMURUGAN U

ABSTRACT

The rapid advancement of deep learning and Generative Adversarial Networks (GANs) has led to the emergence of highly realistic manipulated images and videos known as deepfakes. These synthetic media pose serious ethical, security, and social threats by enabling the creation of deceptive content that is nearly indistinguishable from authentic visuals. As a result, the detection of deepfakes has become a critical research domain in computer vision and multimedia forensics. This study presents a comprehensive comparative analysis of existing deepfake image detection models developed between 2018 and 2025, emphasizing their methodologies, performance, and limitations. The comparison encompasses three major categories of detection techniques: handcrafted feature-based models such as Local Binary Pattern (LBP) and Co-occurrence Matrix (GLCM), deep learning-based architectures including MesoNet, XceptionNet, and EfficientNet, and hybrid frameworks that integrate both statistical and convolutional features. Each method is evaluated based on its feature extraction approach, dataset dependency, computational complexity, and overall accuracy. Findings from the reviewed literature reveal that handcrafted models provide interpretability and efficiency but lack scalability, while deep learning systems achieve high accuracy at the cost of computational resources. The study concludes that hybrid frameworks—particularly those combining Co-occurrence Matrix and Convolutional Neural Networks (CNN) offer the most balanced trade-off between performance, generalization, and interpretability. These models consistently achieve accuracies exceeding 89%, outperforming traditional handcrafted approaches and demonstrating robustness under varying compression and lighting conditions. This comparative study serves as a consolidated reference for future research, providing key insights into the evolution, effectiveness, and challenges of deepfake image detection methodologies.

Keywords — Deepfake detection, feature extraction, image forensics, convolutional neural networks, co-occurrence matrix, transformer models, texture analysis, frequency-domain features, hybrid detection methods, generative adversarial networks, comparative analysis, image manipulation, digital forensics, detection accuracy, deep learning approaches, dataset analysis.

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ABBREVIATIONS

- AI – Artificial Intelligence
- API – Application Programming Interface
- AUC – Area Under Curve
- CNN – Convolutional Neural Network
- CPU – Central Processing Unit
- DL – Deep Learning
- DF – Deepfake
- DFDC – DeepFake Detection Challenge
- DFT – Discrete Fourier Transform
- DNN – Deep Neural Network
- FF++ – FaceForensics++
- GAN – Generative Adversarial Network
- GLCM – Gray Level Co-occurrence Matrix
- GPU – Graphics Processing Unit
- LBP – Local Binary Pattern
- ML – Machine Learning
- MLP – Multi-Layer Perceptron
- PCA – Principal Component Analysis
- PSNR – Peak Signal-to-Noise Ratio
- ReLU – Rectified Linear Unit
- ROC – Receiver Operating Characteristic
- ROC-AUC – Receiver Operating Characteristic - Area Under Curve
- SVM – Support Vector Machine